

Annexure No.:

## EXPERIMENT - 11

Aim: Demonstrate the working of Ultrasonic Distance Measurement sensor.

Apparatus:

ultrasonic , sensor , Arduino board , USB cable , connecting wires.

Theory:

The ultrasonic sensor or HC-SR04 is used to measure the distance of the object using SONAR.

It emits the ultrasound at a frequency of 40KHZ or 40000 Hz. The frequency travels through the air and strikes the objects on its path. The rays, bounce back from the object and reach back to the module.

- The ultrasonic sensor works best for medium ranges.
- The resolution is 0.3 cm.
- The medium ranges of the sensor are 10 cm to 3 cm. It works best at this duration.
- The maximum range the sensor may detect is 4.5 m.

## Performance Result:



1) The distance of the object is measured by the sensor. The sensor is connected to the Arduino board. The sensor is connected to the Arduino board. The sensor is connected to the Arduino board.

2) The distance of the object is measured by the sensor. The sensor is connected to the Arduino board. The sensor is connected to the Arduino board. The sensor is connected to the Arduino board.

Conclusion:

The sensor is connected to the Arduino board.

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Apparatus:

Distance measurement sensor

Arduino Uno board, breadboard, jumper wires, power supply, etc.

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CODE :

```
const int trigPin = 9;
const int echoPin = 10;
long duration;
int distance;
void setup() {
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    serial.begin(9600);
}
void loop() {
    digitalWrite(trigPin, LOW);
    delayMicrosecond(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;
    serial.print("Distance :");
    serial.print(distance);
    serial.println("cm");
    delay(500);
}
```

conclusion :

The HC-SR04 ultrasonic sensor with Arduino Uno accurately measure distance by sending and receiving ultrasonic waves making it useful for obstacle detection and distance measurement application.